ModelIt!

Student Activity Pack

Why Does Your Bike Rust?

How Water and Oxygen Change Metal Over Time

Grade 04L2 | NGSS 4-PS1-2, 4-ETS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever seen rust on a bike, car, or piece of metal? What did it look like?**

2. **What do you think causes metal to rust? What conditions make rust happen?**

3. **Draw what you think is happening to the metal when it turns into rust. Is it just changing color, or is something else happening?**

4. **If you wanted to stop a piece of metal from rusting, what would you do? Why do you think that would work?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Exposure | Oxygen Level | Oxidation Rate | Rust Formation | Metal Strength

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Rainy and Open Air

Water Exposure at 80%, Oxygen Level at 80%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Dry and Open Air

Water Exposure at 20%, Oxygen Level at 80%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Wet but Sealed

Water Exposure at 80%, Oxygen Level at 20%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Salt Presence — Whether salt is in the water or air. Salt water causes metal to rust much faster than fresh water because salt helps the chemical reaction happen more easily

Temperature — How warm or cold the environment is. Warmer temperatures speed up chemical reactions, including rusting, while very cold temperatures slow them down

Metal Type — What kind of metal the object is made from. Iron and steel rust easily, while aluminum, copper, and stainless steel resist corrosion through different chemical properties

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If your bike is made of strong metal, why does leaving it outside in the rain turn that strong metal into crumbly, reddish-brown rust? Use evidence from your simulation to support your answers.

1 Why does a bike left in the rain rust faster than one kept in a dry garage?

2 Is rust just a color change, or is something more happening to the metal? How do you know?

3 Why do you think painting metal helps prevent rust?

4 If rust needs both water and oxygen, would a nail at the bottom of a sealed jar of water rust? Why or why not?

5 How did your model help you understand why some conditions cause more rust than others?

8 STEM Challenge: Design the Rust-Proof Bike

ModelIt!

YOUR ENGINEERING MISSION

Design a protective system that prevents a steel nail from rusting when exposed to water and air for one week.

THE SCENARIO

A bicycle company has a problem: customers in rainy cities are complaining that their bikes are rusting within months. The company needs your engineering team to test different protective coatings and recommend the best way to keep bikes rust-free. You must test at least three different coatings on steel nails, expose them to water for one week, and present your findings with evidence about which coating works best and why.

Engineering Design Thinking

1. WHAT COATINGS WILL YOU TEST (PETROLEUM JELLY, NAIL POLISH, COOKING OIL, WAX, TAPE, ALUMINUM FOIL)?
2. HOW WILL YOU MAKE SURE YOUR TEST IS FAIR — SAME TYPE OF NAIL, SAME AMOUNT OF WATER, SAME AMOUNT OF TIME?
3. HOW WILL YOU MEASURE HOW MUCH RUST FORMS ON EACH NAIL AFTER ONE WEEK?
4. WILL YOU ALSO TEST ONE NAIL WITH NO COATING AS A CONTROL TO COMPARE AGAINST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials Scientists study how different substances behave and develop new materials that resist damage from rust, heat, and wear. They create the coatings, alloys, and composites used in everything

from bridges to spacecraft. They earn $75,000-$120,000/year.